Pinecone Weather Machine

Build a weather sensor with no battery and no code.

Win condition: Most accurate, readable, and fast response after spray-and-dry trials.

THE SCIENCE YOU ARE USING

Hygromorphs. A pine cone opens in dry air and closes when it is humid. Its scales are built from two layers that swell by different amounts when they absorb moisture, so the scale bends. No motor, no power, just material that responds to water in the air.

Bilayer biomimicry. You can copy this with a paper-and-tape bilayer. One layer absorbs moisture and grows; the other does not. The mismatch makes your indicator curl or straighten. The goal is a readable, repeatable humidity signal.

HOW TO BUILD AND RUN IT

01 **Observe a real cone.** Spray a pine cone and watch the scales close. Let it dry and watch them open.

02 **Build a bilayer indicator.** Tape a moisture-absorbing layer to a stiff backing so the strip bends when wet.

03 **Add a readable scale.** Mount the strip against a ruler or dial so the bend points to a number.

04 **Run spray-and-dry trials.** Spray, time the response, dry, repeat. Record how far and how fast it moves.

05 **Redesign for speed.** Adjust layer thickness or length to respond faster and more clearly.

Safety: Spray water only at the one test spot and wipe up right away so the floor stays dry and no one slips. Inspect cones for sap, insects, and sharp broken scales. Handle skewer points carefully. Allergy: if you are sensitive to pine, use the paper-only bilayer instead of handling a cone. No goggles or gloves are needed.

MATERIALS

Dry pine cones	shared
Index cards	shared
Wood skewers	shared
Spray bottles	1 per team (4)
Rulers	1 per team (4)
Tape and fasteners	shared

YOUR PLAN AND DATA

Clean, complete data is worth as much as a fast result.

Prediction (what will work best, and why): ____________________

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The ONE variable we are testing (strip length, paper-layer thickness, OR spray pumps): ____________________

Baseline spray-and-dry trial: tip moved ______ (units on ruler), time to respond ______ s

After redesign: tip moved ______ (units on ruler), time to respond ______ s

DEFEND YOUR DESIGN

What evidence made you change your design? ____________________

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If you ran it again, what is the ONE thing you would change? ____________________

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HOW YOU SCORE (100 POINTS)

SCORING CRITERION	POINTS
Response accuracy	35
Response speed	20
Readable output scale	20
Biomimicry explanation	15
Redesign quality	10
Total	**100**